Proficiency Testing (PT) Participation Plan (2025 – 2029)

Laboratory Name: Azuresky Pty Ltd

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1. Objective

To define the laboratory's 5-year strategy for participation in internal and external proficiency testing schemes. The plan ensures continuous verification of analytical competence for coal-related parameters under controlled, low-throughput conditions where PT complements other quality-assurance (QA) measures.

2. Scope

Parameter	Method/Standard
Quick Ash	In-house method (based on ISO 1171)
ISO Ash	ISO 1171
ISO Volatile Matter	ISO 562
ISO Moisture (Inherent Moisture)	ISO 589 / SANS 5925
Calorific Value	ISO 1928

3. Laboratory Context

Operational focus: Quality assurance and validation of homogenised reference coal materials. Testing frequency is occasional, primarily for CRM stability checks and verification of homogenised lots. Duplicate analyses are performed for each batch, accompanied by CRMs to monitor drifts.

Internal QC practices include weekly verification of crushers, furnaces, and ovens (every Monday). Furnace and oven calibration is externally done by Micron, while the calorimeter is calibrated in-house. External technicians service crushers and pulverisers.

4. Quality-Assurance Measures Supporting PT

QA / IQC Measure	Frequency	Purpose
CRM analysis with each duplicate batch	Each testing event	Detect and quantify drift or bias
Duplicate sample analysis	Each testing event	Check repeatability
Crusher verification	Weekly (Monday)	Validate sample-preparation uniformity
Furnace & oven verification	Weekly (Monday)	Confirm temperature accuracy
CRM stability check	Monthly	Detect degradation of stored reference materials
External calibration (Micron)	Annual	Ensure metrological traceability
External servicing of mechanical units	As scheduled	Maintain performance stability

5. Risk-Based Determination of PT Frequency

Risk Factor	Risk Level	Mitigation / QA Action
Limited sample throughput	Low	Monthly CRM verification plus quarterly PT participation
Staff turnover	Low	Cross-training and annual competency review
Instrument drift	Moderate	Duplicate,CRM monitoring, weekly verifications
Method complexity	Moderate	Periodic bias trending and external PT
External PT availability	Readily available	Enrolled quarterly

6. Areas of Technical Competence

Technical Area	Measurement Process	Parameters Included	Key Equipment
Proximate Analysis	Gravimetric/ Thermogravimetric	Quick Ash, ISO Ash, ISO Volatile Matter, ISO Moisture	Muffle furnace, drying oven
Calorific-Value Determination	Combustion Calorimetry	Calorific Value	Bomb calorimeter

7. PT Participation Strategy and Frequency

Parameter	External PT (CoalConcept)	Internal PT (CRM Checks)	Total Frequency	Rationale
Quick Ash	Quarterly	Monthly CRM check + duplicates	4 external + 12 internal	Moderate risk; temperature-dependent
ISO Ash	Quarterly	Monthly CRM check	4 external + 12 internal	Linked to calorific correlation
ISO Volatile Matter	Quarterly	Monthly CRM check	4 external + 12 internal	Sensitive to furnace behaviour
ISO Moisture	Quarterly	Monthly CRM check	4 external + 12 internal	Influenced by humidity
Calorific Value	Quarterly	Monthly CRM check	4 external + 12 internal	Key coal quality indicator

8. Graphical Summary – PT Cycle

The diagram below illustrates the PT cycle and review process:

Weekly Equipment Verification → Monthly CRM/Duplicate Check → Quarterly External PT (CoalConcept) → Bias & Trend Evaluation → Corrective Actions if Needed → Annual Management Review → PT Plan Update

9. Conclusion

This PT Plan demonstrates a sound, risk-based approach suited to a non-commercial coal reference laboratory with limited sample throughput. Through structured internal CRM verification, weekly equipment checks, and quarterly external PT participation, the laboratory ensures ongoing validation of analytical competence and readiness for SANAS accreditation.